

Seafood Fraud Social Food Web Model

The below deterministic discrete-time Consumer-Resource model is inspired by those of Fryxell et al. (2010), McCann et al. (2016), Fryxell et al. (2017), Bieg et al. (2017), and Bieg and McCann (2020), incorporating elements of fisheries bioeconomic theory, community ecology theory, and human behavioural ecology theory.

The Resource (Seafood biomass; S) follows a Ricker logistic surplus production dynamics model of net recruitment subject to density dependence (through maximum per capita exponential growth rate ($r_{\max,S}$) and linear harvest (*i.e.*, catch) (H_S)). Harvest comes from two types of Consumers: Buyers (B), who seek to purchase a proportion (f) of legally-caught and correctly-labelled seafood at market price, as well as Fraudsters (F), who seek to illicitly misrepresent the remaining fraction ($1 - f$) of illegally-caught seafood through adulteration, mislabelling, substitution, or other forms of fraud. In this sense, Seafood can be viewed as producers, Fraudsters as predators, and Buyers as prey, interacting with one another in a social meta food web (*i.e.*, the seafood supply chain) (Bieg et al., 2022) linked by smaller food webs (*i.e.*, fisheries) that exchange information (Bieg et al., 2018). Buyers and Fraudsters compete with one another for the Resource and their levels are regulated through their own behaviours and profits. Harvest of seafood depends on fishing effort/intensity (E) expended to obtain seafood (*e.g.*, the number of boats, the number of fishers, the number of meters of gillnet, the number of hooks set, *etc.*) scaled by catchability (q) of the seafood (the ability of Buyers and Fraudsters to obtain the resource for every one unit of expended effort). Since Buyers and Fraudsters make up the entire human population with the system, their levels are conserved; thus, their levels at time t (as proportions of their carrying capacities, which are set to one) must sum to one (*i.e.*, $B_t + F_t = 1$). Therefore,

$$B'_t = \frac{B_t}{B_t + F_t} \quad (1)$$

$$F'_t = \frac{F_t}{B_t + F_t} = 1 - B'_t. \quad (2)$$

Harvest of Seafood by Buyers and Fraudsters, respectively, is given by

$$H_{B,t} = q_B E_{B,t} S_t \quad (3)$$

$$H_{F,t} = q_F E_{F,t} S_t \quad (4)$$

where harvesting Effort by Buyers and Fraudsters, respectively, takes the form

$$E_{B,t} = f B_t \quad (5)$$

$$E_{F,t} = (1 - f) F_t \quad (6)$$

so that,

$$H_{B,t} = q_B f B_t S_t \quad (7)$$

$$H_{F,t} = q_F (1 - f) F_t S_t \quad (8)$$

where the total Harvest of Seafood is

$$H_{S,t} = H_{B,t} + H_{F,t} \quad (9)$$

$$= q_B f B_t S_t + q_F (1 - f) F_t S_t. \quad (10)$$

Therefore, the Seafood equation is

$$S_{t+1} = S_t e^{r_{\max, S}(1-S_t)} - q_B f B_t S_t - q_F (1-f) F_t S_t. \quad (11)$$

Note, the Catch-per-unit-Effort (CPUE) by Buyers and Fraudsters, respectively, is defined as

$$\text{CPUE}_{B,t} = q_B S_t \quad (12)$$

$$\text{CPUE}_{F,t} = q_F S_t \quad (13)$$

and the fishing “mortality” (*i.e.*, harvest/catch divided by biomass) due to purcurment of seafood by Buyers and Fraudsters, respectively, is

$$m_{B,t} = q_B B_t \quad (14)$$

$$m_{F,t} = q_F F_t. \quad (15)$$

Equation (11) can thus be expressed as

$$S_{t+1} = S_t e^{r_{\max, S}(1-S_t)} - f B_t \text{CPUE}_{B,t} - (1-f) F_t \text{CPUE}_{F,t} \quad (16)$$

$$= S_t e^{r_{\max, S}(1-S_t)} - f m_{B,t} S_t - (1-f) m_{F,t} S_t. \quad (17)$$

Fraudsters become more irrational as the difference between market seafood price and wholesale seafood price increases ($\Delta P = P_{\text{Market}} - P_{\text{Wholesale}}$). Typically, market seafood prices are higher than wholesale prices due to the added processing of seafood required prior to the point of sale. When $\Delta P < 0$, Fraudsters are negatively impacted, decreasing their

levels. In the case of $\Delta P > 0$, Fraudsters benefit, increasing their levels. If $\Delta P = 0$, Fraudster levels remain unchanged.

Buyers and Fraudsters seek to optimize their overall behaviours and profits in obtaining Seafood. Profit is the difference between revenue of the seafood (landed price times harvest) and total cost of the seafood (cost per unit effort times effort). That is,

$$\Pi_{B,t} = \pi_B q_B f B_t S_t - c_B f B_t \quad (18)$$

$$= \pi_B f B_t \text{CPUE}_{B,t} - c_B f B_t \quad (19)$$

$$= \pi_B f m_{B,t} S_t - c_B f B_t \quad (20)$$

$$\Pi_{F,t} = \pi_F q_F (1-f) F_t S_t - c_F (1-f) F_t \quad (21)$$

$$= \pi_F (1-f) F_t \text{CPUE}_{B,t} - c_F (1-f) F_t \quad (22)$$

$$= \pi_F (1-f) m_{F,t} S_t - c_F (1-f) F_t. \quad (23)$$

The Buyer equation is thus given by

$$B_{t+1} = B_t e^{\alpha B_t S_t - \beta B_t F_t + \pi_B q_B f B_t S_t - c_B f B_t} \quad (24)$$

$$= B_t e^{\alpha B_t S_t - \beta B_t F_t + \pi_B f B_t \text{CPUE}_{B,t} - c_B f B_t} \quad (25)$$

$$= B_t e^{\alpha B_t S_t - \beta B_t F_t + \pi_B f m_{B,t} S_t - c_B f B_t}. \quad (26)$$

The Fraudster equation takes the form

$$F_{t+1} = F_t e^{(\gamma F_t S_t - \delta F_t B_t) \epsilon \Delta P + \pi_F q_F (1-f) F_t S_t - c_F (1-f) F_t} \quad (27)$$

$$= F_t e^{(\gamma F_t S_t - \delta F_t B_t) \epsilon \Delta P + \pi_F (1-f) F_t \text{CPUE}_{B,t} - c_F (1-f) F_t} \quad (28)$$

$$= F_t e^{(\gamma F_t S_t - \delta F_t B_t) \epsilon \Delta P + \pi_F (1-f) m_{F,t} S_t - c_F (1-f) F_t}. \quad (29)$$

The parameters α , β , γ , and δ represent the degree of awareness of Buyers to changing levels of seafood, the degree of awareness of Buyers to evolving Fraudster behaviour, the degree of awareness of Fraudsters to shifting seafood levels, and the degree of awareness of Fraudsters to altering Buyer behaviour, respectively. The parameter ϵ reflects the degree of awareness of Fraudsters to price volatility.

The quantities $\alpha B_t S_t - \beta B_t F_t$ and $(\gamma F_t S_t - \delta F_t B_t)\epsilon \Delta P$ in Equations (24)-(26) and Equations (27)-(29) can be thought of as behaviour terms for Buyers and Fraudsters, respectively. The quantities $\pi_B q_B f B_t S_t - c_B f B_t$ and $\pi_F q_F (1 - f) F_t B_t - c_F (1 - f) F_t$ within the same equations are profitability terms, where π_B and π_F are prices for seafood paid by Buyers and Fraudsters, respectively, and c_B and c_F are costs per unit effort expended by Buyers and Fraudsters to obtain seafood, respectively.

The final model contains three variables indexed by time (S_t , B_t , and F_t), 14 fixed parameters ($r_{\max,S}$, f , q_B , q_F , π_B , π_F , c_B , c_F , α , β , γ , δ , ϵ , and ΔP), and four derived quantities ($\text{CPUE}_{B,t}$, $\text{CPUE}_{F,t}$, $m_{B,t}$, and $m_{F,t}$). Due to the nonlinear nature of the model, all parameters except $r_{\max,S}$ (> 0), and ΔP are constrained to $[0, 1]$ to avoid very large, very small, or infinite values of the variables that could arise due to the exponential terms within the model.

Equilibria and Stability

The system has eight equilibria (with many special cases):

$$(S^*, B^*, F^*) = (0, 1, 0) \quad (30)$$

$$(S^*, B^*, F^*) = (0, 0, 1) \quad (31)$$

$$(S^*, B^*, F^*) = (1, 1, 0) \quad (32)$$

$$(S^*, B^*, F^*) = (1, 0, 1) \quad (33)$$

$$(S^*, B^*, F^*) = \left(1 - \frac{1}{r_{\max,S}} \ln(1 + q_B f), 1, 0\right) \quad (34)$$

$$(S^*, B^*, F^*) = \left(1 - \frac{1}{r_{\max,S}} \ln(1 + q_F(1 - f)), 0, 1\right) \quad (35)$$

$$(S^*, B^*, F^*) = \left(\frac{\beta + c_B f}{\alpha + \pi_B q_B f}, 0, 1\right) \quad (36)$$

$$(S^*, B^*, F^*) = \left(\frac{\delta \epsilon \Delta P + c_F(1 - f)}{\gamma \epsilon \Delta P + \pi_F q_F(1 - f)}, 1, 0\right). \quad (37)$$

When $q_B = 0$, $f = 0$, or both, with $r_{\max,S} > 0$ in all cases, Equation (34) becomes

$$(S^*, B^*, F^*) = (1, 1, 0). \quad (38)$$

When both $q_B = 1$ and $f = 1$, with $r_{\max,S} > 0$ in all cases, Equation (34) becomes

$$(S^*, B^*, F^*) = (1 - \ln(2), 1, 0) \approx (0.307, 1, 0). \quad (39)$$

When $q_F = 1$ and $f = 0$, with $r_{\max,S} = 1$ in all cases, Equation (35) becomes

$$(S^*, B^*, F^*) = (1 - \ln(2), 0, 1) \approx (0.307, 0, 1). \quad (40)$$

Similarly, when $q_F = 0$ and $f = 1$, with $r_{\max,S} = 1$ in all cases, Equation (35) becomes

$$(S^*, B^*, F^*) = (1 - \ln(2), 0, 1) \approx (0.307, 0, 1). \quad (41)$$

When $f = 0$, Equation (36) becomes

$$(S^*, B^*, F^*) = \left(\frac{\beta}{\alpha}, 0, 1 \right), \quad (42)$$

with the added condition that $\alpha \geq \beta$ to ensure $S^* \in [0, 1]$.

When both $\alpha = 0$ and $\beta = 0$, Equation (36) becomes

$$(S^*, B^*, F^*) = \left(\frac{c_B}{\pi_B q_B}, 0, 1 \right), \quad (43)$$

for any value of $0 < f \leq 1$, with the additional constraint that $c_B \leq \pi_B q_B$ to ensure $S^* \in [0, 1]$.

When $f = 1$, Equation (37) becomes

$$(S^*, B^*, F^*) = \left(\frac{\delta}{\gamma}, 1, 0 \right), \quad (44)$$

for any value of $\epsilon \neq 0$ and $\Delta P \neq 0$, with the additional constraint that $\delta \leq \gamma$ to ensure $S^* \in [0, 1]$.

When one or more of γ , δ , ϵ , and ΔP is zero, Equation (37) reduces to

$$(S^*, B^*, F^*) = \left(\frac{c_F}{\pi_F q_F}, 1, 0 \right), \quad (45)$$

for any value of $0 \leq f < 1$, with the additional constraint that $c_F \leq \pi_F q_F$ to ensure $S^* \in [0, 1]$.

From Equations (36) and (37), a sharper bound on f is

$$f \in \left[\frac{\beta - \alpha}{\pi_B q_B - c_B}, \frac{(\gamma - \delta)\epsilon}{\pi_F q_F - c_F} \Delta P + 1 \right] \quad (46)$$

which reduces to $f \in [0, 1]$ (which is the case for Equations (32) and (33)) when $\alpha = \beta$ and either $\gamma = \delta$, $\epsilon = 0$, or $\Delta P = 0$.

Are there any biologically meaningful combinations of variables/parameters leading to

oscillations, bifurcations, limit cycles, or chaos?

Model Extensions

- Add interaction term between Seafood, Buyers and Fraudsters to Buyer and Fraudster equations – $\eta B_t F_t S_t$ (not sure of the sign, positive or negative?)
- Wild *vs.* farmed fish
- Ethical harvesters (*e.g.*, Indigenous communities)
- Legislators/policymakers/fisheries managers
- Agent-based model - Seafood, Buyer, and Fraudster levels are updated according to simple deterministic/stochastic rules
- Game theory - Buyers and Fraudsters play a zero-sum game as they compete for Seafood, whereby one player's gain is the other's loss

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